Mathematics in the Modern World

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Propositional Logic

Introduction:

Logic

- is the study of the logic relationships between objects and
- forms the basis of all mathematical reasoning and all automated reasoning.

Introduction:

In **Propositional Logic**

- (a.k.a Propositional Calculus or Sentential Logic)

- the objects are called **propositions.**

Definition:

A **proposition**

- is a statement that is either true or false, but not both.
- We usually denote a proposition by a letter: *p*, *q*, *r*, *s*, …

Definition:

- The value of a proposition is called its **truth value**; denoted by
  - ***T* or 1 if it is true or**
  - ***F* or 0 if it is false**

- Opinions, interrogative, and imperative are not propositions.

Truth table

p
0
1

Propositions: Examples

- The following are propositions.
  - Today is Tuesday.
  - The grass is wet.
  - It is raining.
- The following are not propositions
  - C++ is the best language. ***Opinion***
  - When is the pretest? ***Interrogative***
  - Do your homework. ***Imperative***

Are these propositions?

- 2+2=5
- Every integer is divisible by 12.
- Microsoft is an excellent company.

Logical Connectives

- Connectives are used to create a compound proposition from two or more propositions.
  - Negation (denote $\neg$)
  - And or logical conjunction (denoted $\wedge$)
  - Or or logical disjunction (denoted $\vee$)
  - XOR or exclusive or (denoted $\oplus$)
  - Implication (denoted $\Rightarrow$ or $\rightarrow$)
  - Biconditional (denoted $\Leftrightarrow$ or $\leftrightarrow$)

- We define the meaning (semantics) of the logical connectives using **truth tables**.

Logical Connective: Negation

- $\neg p$, the negation of a proposition p, is also a proposition.

- Examples:
  - Today is not Monday.
  - It is not the case that today is Monday, etc.

Truth table

p	$\neg p$
0	1
1	0

Logical Connective: Logical And

- The logical connective And is true only when both of the propositions are true.
- It is also called a conjunction.

- Examples
  - It is raining and it is warm.
  - (2+3=5) and (1<2)

Truth table

p	q	$p \wedge q$
0	0	0
0	1	0
1	0	0
1	1	1

Logical Connective: Logical Or

- The logical disjunction, or logical Or, is true if one or both of the propositions are true.
- Examples
  - It is raining or it is the second lecture.
  - (2+2=5) ∨ (1<2)
  - You may have cake or ice cream.

Truth Table

p	q	$p \wedge q$	$p \vee q$
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1

Logical Connective: Exclusive Or

- The exclusive Or, or XOR, of two propositions is true when exactly one of the propositions is true and the other one is false.

- Example
  - The circuit is either ON or OFF but not both.
  - Let $ab<0$, then either $a<0$ or $b<0$ but not both.
  - You may have cake or ice cream, but not both.

Truth table

p	q	$p \wedge q$	$p \vee q$	$p \oplus q$
0	0	0	0	0
0	1	0	1	1
1	0	0	1	1
1	1	1	1	0

Logical Connective: Implication (1)

- **Definition:** Let p and q be two propositions.
- The implication $p \rightarrow q$ is the proposition that is false when p is true and q is false and true otherwise.
  - p is called the hypothesis, antecedent, premise.
  - q is called the conclusion, consequence.

Truth table

p	q	$p \wedge q$	$p \vee q$	$p \oplus q$	$p \Rightarrow q$
0	0	0	0	0	1
0	1	0	1	1	1
1	0	0	1	1	0
1	1	1	1	0	1

Logical Connective: Implication (2)

- The implication of $p \rightarrow q$ can be also read as
  - If p then q
  - p implies q
  - If p, q
  - p only if q
  - q if p
  - q when p
  - q whenever p
  - q follows from p
  - p is a sufficient condition for q (p is sufficient for q)
  - q is a necessary condition for p (q is necessary for p)

Logical Connective: Implication (3)

- **Examples**
  - If you buy air ticket in advance, it is cheaper.
  - If x is an integer, then $x^2 \geq 0$.
  - If it rains, the grass gets wet.
  - If the sprinklers operate, the grass gets wet.
  - If 2+2=5, then all unicorns are pink.

Exercise: Which of the following implications is true?

- If 1 is a positive number, then 2+2=4

  True. Same as above.

- If $\sin x = 0$, then $x = 0$

  False.

  x can be a multiple of π. If we let $x=2\pi$, then $\sin x = 0$ but $x \neq 0$.

  The implication "if $\sin x = 0$, then $x = k\pi$, for some k" is true.

Logical Connective: Biconditional (1)

- **Definition:** The biconditional $p \leftrightarrow q$ is the proposition that is true when p and q have the same truth values. It is false otherwise.
- Note that it is equivalent to $(p \rightarrow q) \wedge (q \rightarrow p)$.

Truth Table

p	q	$p \wedge q$	$p \vee q$	$p \oplus q$	$p \Rightarrow q$	$p \Leftrightarrow q$
0	0	0	0	0	1	1
0	1	0	1	1	1	0
1	0	0	1	1	0	0
1	1	1	1	0	1	1

Logical Connective: Biconditional (2)

- The biconditional $p \leftrightarrow q$ can be equivalently read as
  - p if and only if q
  - p is a necessary and sufficient condition for q
  - if p then q, and conversely
  - p iff q
- Examples
  - $x>0$ if and only if x^2 is positive.
  - The alarm goes off **iff** a burglar breaks in.
  - You may have pudding **iff** you eat your meat.

Exercise: Which of the following biconditionals is true?

- $x^2 + y^2 = 0$ if and only if $x=0$ and $y=0$.

  True. Both implications hold.

- $2 + 2 = 4$ if and only if $\sqrt{2}<2$.

  True. Both implications hold.

- $x^2 \geq 0$ if and only if $x \geq 0$.

  False.
  The implication "if $x \geq 0$ then $x^2 \geq 0$" holds.
  However, the implication "if $x^2 \geq 0$ then $x \geq 0$" is false.

  Consider $x=-1$.
  The hypothesis $(-1)^2=1 \geq 0$ but the conclusion fails.

Truth Tables

- Truth tables are used to show/define the relationships between the truth values of
  - the individual propositions and
  - the compound propositions based on them.

p	q	$p \wedge q$	$p \vee q$	$p \oplus q$	$p \Rightarrow q$	$p \Leftrightarrow q$
0	0	0	0	0	1	1
0	1	0	1	1	1	0
1	0	0	1	1	0	0
1	1	1	1	0	1	1

Constructing Truth Tables

- Construct the truth table for the following compound proposition.

$$((p \wedge q) \vee \neg q)$$

p	q	$p \wedge q$	$\neg q$	$((p \wedge q) \vee \neg q)$
0	0	0	1	1
0	1	0	0	0
1	0	0	1	1
1	1	1	0	1

Precedence of Logical Operators

- As in arithmetic, an ordering is imposed on the use of logical operators in compound propositions.
- However, it is preferable to use parentheses to disambiguate operators and facilitate readability.

$$\neg p \vee q \wedge \neg r \quad \equiv \quad (\neg p) \vee (q \wedge (\neg r))$$

- To avoid unnecessary parenthesis, the following precedences hold:
  1. Negation ($\neg$)
  2. Conjunction ($\wedge$)
  3. Disjunction ($\vee$)
  4. Implication/Conditional ($\rightarrow$)
  5. Biconditional ($\leftrightarrow$)

Usefulness of Logic

- Logic is more precise than natural language.
  - You may have cake or ice cream.
    - Can I have both?
  - If you buy your air ticket in advance, it is cheaper.
    - Are there or not cheap last-minute tickets?

- For this reason, logic is used for hardware and software specification.
  - Given a set of logic statements,
  - One can decide whether or not they are satisfiable (i.e., consistent), although this is a costly process…